

Mohammad Nour

📍 Worcester, MA | 📞 774-670-7742 | ✉️ snouraniboosjin@wpi.edu | 🌐 Mohammad Nour

PROFESSIONAL SUMMARY

Third-year PhD researcher with 5 years of dedicated experience in hardware security and the application of machine learning to cryptographic analysis. Specialized in side-channel analysis with hands-on expertise in training deep learning models for key recovery, developing explainable-AI methods to interpret model behavior, and designing advanced metrics to maximize attack performance. Extensive practical experience with modern machine learning libraries and frameworks (e.g., PyTorch, TensorFlow, scikit-learn) for large-scale experimentation. Strong foundation in cryptography, information theory, and theoretical machine learning, combined with a proven ability to collaborate effectively within multidisciplinary research teams.

SKILLS

Programming Languages: Verilog, Python, C/C++, MATLAB, Shell scripting, solidity.

DL/ML: PyTorch, PyTorch Lightning; TensorFlow (v1/v2); OpenCV; scikit-learn; NumPy, Pandas.

Platforms & Tooling: Debian/GNU Linux, Windows; Git, Docker, bash; basic CUDA.

PROJECTS

Uncertainty Estimation in Neural Network-enabled Side-channel Analysis

CHES 2026

- Developed a framework that quantifies predictive uncertainty in neural-network side-channel attacks, separating contributions from model behavior and data quality.
- Integrated SHAP-based explainability and matrix-based Rényi entropy to reveal which trace features drive uncertainty and influence key-recovery performance.

There's Waldo: PCB Tamper Forensic Analysis using Explainable AI on Impedance Signatures

EMC 2025

- Developed an explainable machine-learning framework that uses impedance (S-parameter) signatures to detect and classify PCB tampering with 96% accuracy.
- Applied SHAP-based feature attribution to identify responsible frequency components, enabling fast, non-invasive hardware forensics.

Optimist: Acceleration of AI for Implementation Security Testing

NSF project

- Contributed to the OPTIMIST initiative by analyzing AI-based acceleration methods for side-channel and fault-attack security testing, focusing on tooling, datasets, and evaluation practices. Identified key gaps in model portability, dataset standardization, and algorithmic guidelines to improve the reliability and reproducibility of AI-driven implementation-security assessments.

Too hot to be true: Temperature calibration for higher confidence in NN-assisted side-channel analysis

CASCADE 2026

- Studied the effects of temperature calibration on deep learning based SCA.
- Introduced matrix-based Rényi entropy and maximum mean discrepancy (MMD) to quantify hardness and explain classification difficulty using SHAP-driven interpretability.

Swarm in EM Hay: Particle Swarm-guided Probe Placement for EM SCA

WOOT 2026

- Studied the effects of temperature calibration on deep learning based SCA.
- Introduced matrix-based Rényi entropy and maximum mean discrepancy (MMD) to quantify hardness and explain classification difficulty using SHAP-driven interpretability.

AWARDS AND HONORS

- Recipient of the **DAC Young Fellow Award 2025**.
- Invited talk at **New England Hardware Security Day 2025**, MIT.
- **3rd Place Best Poster Award** at New England Hardware Security Day 2024, WPI.

EDUCATION

Worcester Polytechnic Institute

PhD - Electrical and Computer Engineering | Advisor: Prof. Fatemeh Ganji

Courses: *Applied cryptography & physical attacks, Physical Security of Microelectronic Systems*

2023 - Present

Worcester, MA

Sharif University of Technology

M.Sc. - secure communications and cryptography

Courses: *Deep learning, Cryptography, foundations of the blockchains*

2020 - 2023